In this paper, all problem in the silverman will be solves in complete form.

#7.1.9.

Let $C: z(t) = e^{it}$ $(-\pi \le t \le \pi)$. Evaluate

$$\int_C z^2\, dz = \int_{-\pi}^{\pi} e^{2it} \cdot i \cdot e^{it}\, dt = i\int_{-\pi}^{\pi} e^{3it}\, dt = i \cdot \left[\frac{1}{3i} \cdot e^{3it} \Big|_{-\pi}^{\pi} \right] = 0$$

$$\int_C \frac{1}{z}\, dz = \int_{-\pi}^{\pi} \frac{1}{e^{it}} \cdot i \cdot e^{it}\, dt = 2\pi i.$$

$$\int_C \frac{1}{z^2}\, dz = \int_{-\pi}^{\pi} i \cdot \frac{1}{e^{it}}\, dt = i \cdot \left[+i \cdot e^{-it} \Big|_{-\pi}^{\pi} \right] = -1 \cdot 0 = 0.$$

$$\int_C \bar{z}\, dz = \int_{-\pi}^{\pi} [\cos t - i \sin t] \cdot i \cdot e^{it}\, dt = i\int_{-\pi}^{\pi} \cos^2 t + \sin^2 t\, dt = 2\pi i.$$

#8.1.2, a)

Let $p(z)$ be a polynomial of degree n, then

$$\int_{|z|=2} \frac{p(z)}{(z-1)^{n+2}}\, dz = 0$$

Prof. By the Cauchy integral formular,

$$\int_{|z|=2} \frac{p(z)}{(z-1)^{n+2}}\, dz = p^{(n+1)}(1) \cdot \frac{(n+1)!}{2\pi i} = 0.$$

#8.1.2, b)

Given $n, m \in \mathbb{N}$, show that $\int_C \frac{(n-1)!\, e^z}{(z-z_0)^n}\, dz = \int_C \frac{(m-1)!\, e^z}{(z-z_0)^m}\, dz$, where C is along z_0.

Prof. By the Cauchy integral formular,

$$\int_C \frac{(n-1)!\, e^z}{(z-z_0)^n}\, dz = \frac{2\pi i}{(n-1)!} \cdot (n-1)! \cdot e^{z_0} = 2\pi i \cdot z_0.$$

Since the value is independent for n, so proved.

#8.1.3. Evaluate for $C: |z|=3$,

$$\int_C \frac{e^z}{z-2}\, dz = 2\pi i \cdot e^2.$$

$$\int_C \frac{e^z \sin z}{(z-2)^2}\, dz = 2\pi i \cdot (e^2 \sin 2 + e^2 \cdot \cos 2).$$

#8.1.4.

$$\int_{|z|=2} \frac{1}{z^2-1}\,dz = \frac{1}{2}\int_{|z|=2} \frac{1}{z-1} - \frac{1}{z+1}\,dz = 0$$

$$\int_{|z|=3} \frac{z^2+3z-1}{(z-1)(z+2)}\,dz = \frac{1}{3}\int_{|z|=3} \frac{z^2+3z-1}{z-1} - \frac{z^2+3z-1}{z+2}\,dz = \frac{1}{3}\cdot 2\pi i\cdot(6) = 4\pi i$$

8.1.6.

observe that; for $|z|=1$, $\mathrm{Re}\, z = \frac{z+\bar{z}}{2} = \frac{1}{2}\cdot\left(z+\frac{1}{z}\right)$.

So $\int_{|z|=1} (\mathrm{Re}\, z)^2\,dz = \frac{1}{4}\int_{|z|=1} \frac{(z^2+1)^2}{z^2}\,dz = \frac{1}{4}\cdot 2\pi i\cdot 8 = 4\pi i.$

And, for $|z-1|=1$, $z = 1+e^{i\theta}$ $(-\pi \le \theta \le \pi)$, so $\bar{z} = 1+e^{-i\theta}$, therefore

$$\bar{z} = 1+\frac{1}{z-1} = \frac{z}{z-1}.$$

Therefore $\int_{|z-1|=1} (\bar{z})^2\,dz = \int_{|z-1|=1} \frac{z^2}{(z-1)^2}\,dz = 2\pi i\cdot 2 = 4\pi i$

And, for $|z-1|=1$, $\mathrm{Im}\, z = \mathrm{Im}(z-1) = \frac{1}{2i}\cdot(z-\bar{z}) = \frac{1}{2i}\cdot\left(\frac{z^2-2z}{z-1}\right)$, so

$$\int_{|z-1|=1} (\mathrm{Im}\, z)^2\,dz = -\frac{1}{4}\cdot\int_{|z-1|=1} \frac{(z^2-2z)^2}{(z-1)^2}\,dz = -\frac{1}{4}\cdot 0 = 0.$$

#8.1.11. Evaluate the limits

$$\lim_{z\to 0} \frac{e^{z}-1-z}{z^2}$$

#8.1.12. If f is analytic at $z=z_0$, and $|f^{(n)}(z_0)| \leq n^k$ for each n.
Then f is an entire.
Proof. Since f is analytic at z_0, all derivates $f^{(n)}(z_0)$ exist and by the assumption,

$$f(z)=\sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!}\cdot(z-z_0)^n$$

Converges for any $z\in\mathbb{C}$, So f is analytic.
(More specifically,

$$\limsup_{n\to\infty} \frac{(n+1)^k}{(n+1)!}\cdot\frac{n!}{n^k} = \limsup_{n\to\infty.} \left(1+\frac{1}{n}\right)^k\cdot\frac{1}{n+1} = 0.$$

#8.1.13. Show that the following series are analytic.

$$f(z)=\sum_{n=1}^{\infty}\frac{1}{n^z} \quad (\operatorname{Re} z>1), \qquad f(z)=\sum_{n=1}^{\infty}\frac{1}{z^2-n^2} \quad (z\neq \pm1, \pm2, \cdots)$$

Proof. Recall that; $n^z = e^{z\cdot \ln n}$

$$2^x = e^{x\cdot \ln 2}$$

「Section. 8.2.

#8.2.1. If $f(z)=\sum_{n=0}^{\infty} a_n z^n$ is analytic in $|z|<R$, and $f(x)$ is real when $-R<x<R$. Show that all a_n are real, and $f(\bar{z})=\overline{f(z)}$.

Proof. Let $g(z)=\overline{f(\bar{z})}$. So that

$$g(z)=\sum_{n=0}^{\infty}\overline{a_n}\cdot z^n$$

By the assumption, $f(x)-g(x)=\sum_{n=0}^{\infty}(a_n-\bar{a}_n)\cdot x^n=0$ for all $x\in(-R,R)$,

So $a_n-\bar{a}_n=0$ for all n.

#8.2.2. Let $P(z)=a_n z^n+\cdots+a_0$ $(a_n\neq 0)$.

Given $\varepsilon>0$, there exists $R>0$ such that

$$r>R \Rightarrow (1-\varepsilon)\cdot|a_n|\cdot r^n\le|P(z)|\le(1+\varepsilon)|a_n|\cdot r^n \quad \text{for all } |z|=r.$$

#8.2.3.

If all the zeros of a polynomial $p(z)$ have positive real part, then all the zeros of $p'(z)$ have positive real part.

Proof. Denote that $P(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0 = a_n \cdot (z - z_1) \cdots (z - z_n)$.

Then, $p'(z) = a_n \cdot \left[\sum_{i=1}^{n} \prod_{j \neq i} (z - z_j) \right]$, so therefore

$$\frac{p'(z)}{p(z)} = \sum_{i=1}^{n} \frac{1}{z - z_i} \quad \text{for } z \neq z_1, \cdots, z_n.$$

If $\operatorname{Re} z \leq 0$, then $\operatorname{Re}(z - z_i) < 0$ for all $1 \leq i \leq n$,

so $p'(z)/p(z)$ cannot be zero, so $p'(z)$ cannot be zero.

Therefore, if $p'(z) = 0$, $\operatorname{Re} z > 0$.

#8.2.4.

Let f be an entire satisfying $|f(z)| \leq |e^z|$ for all $z \in \mathbb{C}$.

Show that $f(z) = k e^z$, $|k| \leq 1$.

Proof. Let $g(z) = f(z)/e^z$. Since f is entire, so g is.

And, $|g(z)| = |f(z)| / |e^z| \leq 1$, so $g(z)$ must be a constant, by the Liouville's Thm.

Now, $f(z) = g(z) \cdot e^z$, where g is a constant satisfying $|g| \leq 1$.

#8.2.6.

Let f be an entire s.t. $|f'(z)| \leq |z|$ for all $z \in \mathbb{C}$,

Then $f(z) = az^2 + b$, where $a, b \in \mathbb{C}$ with $|a| \leq \frac{1}{2}$.

Proof. Given $z_0 \in \mathbb{C}$, and $R > 0$, $|f'(z)| \leq |z| = |z - z_0 + z_0| \leq R + |z_0|$, for $|z - z_0| = R$.

So, $$|f''(z_0)| \leq \frac{1!}{R} \cdot \sup_{|z - z_0| = R} f'(z) \leq \frac{R + |z_0|}{R} = 1 + \frac{|z_0|}{R} \to 1 \quad \text{as } R \to \infty.$$

and, for $n \geq 3$,

$$|f^{(n)}(z_0)| \leq \frac{(n-1)!}{R^{n-1}} \cdot \sup f' \leq \frac{R + |z_0|}{R^{n-1}} \to 0 \quad \text{as } R \to \infty.$$

And, $$|f'(z_0)| \leq \frac{1}{2\pi} \cdot \int_C \left| \frac{f(z)}{z} \right| |dz| \leq \frac{2\pi R}{2\pi} \to 0 \quad \text{as } R \to 0,$$

Sec 9.2.

#1. Suppose that $f(z)$ has an iso. sing. at z_0.
If $\lim_{z\to z_0}(z-z_0)^{\alpha}\cdot f(z)=M$ or ∞, where $M\neq 0$, then α must be integer.

proof. Suppose that $\lim_{z\to z_0}(z-z_0)^{\alpha}\cdot f(z)=M$. Then, given $\varepsilon>0$, there is $\delta>0$ s.t
$0<|z-z_0|<\delta \Rightarrow |(z-z_0)^{\alpha}\cdot f(z)|<|M|+\varepsilon$.

So, the $(z-z_0)^{\alpha}\cdot f(z)$ is bounded in the punctured $0<|z-z_0|<\delta$
Then, $(z-z_0)^{\alpha}\cdot f(z)$ has a Taylor Expression

$$(z-z_0)^{\alpha}f(z)=a_0+a_1(z-z_0)+\frac{1}{2!}\cdot a_2(z-z_0)^2+\cdots$$

#2. If f is analytic in $B_R(0)\setminus\{0\}$ and $\lim_{z\to 0}|zf(z)|=0$.
Then the origin is removable.
proof. Since